# SUGGESTED SOLUTIONS TO 2021 H2 PHYSICS PRELIM PAPER 2

Q1(a)

The resistance is given by $R = V/I$. Thus,

$$\frac{\Delta R}{R} \text{x}100 = \left(\frac{\Delta V}{V} + \frac{\Delta I}{I}\right)\text{x}100$$

$$= \left(\frac{0.01}{1.30} + \frac{0.01}{0.76}\right)\text{x}100$$

$\approx$ **2.1 %**

Q1(b)

$$\rho = \frac{RA}{L} = \frac{R\left(\frac{\pi d^2}{4}\right)}{L}$$

$$\rho = \frac{RA}{l} = \frac{\left(\frac{1.30}{0.76}\right)\left(\frac{\pi}{4}\text{x}0.00054^2\right)}{0.754}$$

$$= 5.196\text{x}10^{-7}\ \Omega\,\text{m}$$

$$\frac{\Delta\rho}{\rho} = \frac{\Delta R}{R} + 2\frac{\Delta d}{d} + \frac{\Delta L}{L}$$